WHY DO SOME CUSTOMERS BROWSE BUT NEVER BUY?

A Customer Segmentation Study Using K-Means Clustering

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In One Line

Customers who browse the most spend the least. Some customers visit the website 3x more often than the biggest spenders, yet spend 12x less, a gap this paper investigates and explains.

- 2,211 real customers analysed, grouped into 4 distinct segments using K-Means clustering
- The biggest spenders barely browse; the biggest browsers, on the other hand, barely spend.
- This points to a psychological gap between browsing and buying, and not just a spending gap

THE QUESTION

Most companies assume that a customer who visits their website often is close to buying something. This project tested that assumption using real data from 2,211 customers, and the answer turned out to be more interesting than expected: some of the most frequent browsers were among the lowest spenders, while some of the highest spenders barely visited the website at all.

METHODOLOGY

First and foremost the data was cleaned by dropping rows with missing income, removing implausible values (incomes greater than $200,000, birth years earlier than 1940), leaving 2,211 customers. The raw fields of age, total spend, total purchases, and number of children were created and merged with income, web visits and recency into 7 features that were standardised before clustering. K-Means was run from k=2 to 8; max silhouette scores were at k=2 (0.314) and gradually declined thereafter (k=4: 0.202). k=2 was not used as it reduced the sample into just 2 categories, a low-versus-high spending split with limited managerial value. Instead, the number of clusters was set to 4, which distinguished customers based on both browsing behaviour and spend, while maintaining a reasonable level of statistical fit and interpretability. Full code: Python (pandas, scikit-learn). Dashboard: PowerBI.

THE FOUR CUSTOMER GROUPS

Segment	Who They Are	Avg. Income	Avg. Spend	Avg Web Visits
High Spenders	Older, higher-income, buy often and rarely browse first and fewest number of children	$78,484	$1,390	2.3/month
Window Shoppers	Oldest, lower-income, browse a lot, rarely buy and the most number of children	$41,831	$130	6.0/month

Steady Buyers	Second oldest, mid-high income, balanced browsing and buying and average number of children	$60,877	$859	5.6/month
Budget Browsers	Youngest, lowest-income, highest browsing, lowest spend and few children	$30,268	$112	6.9/month

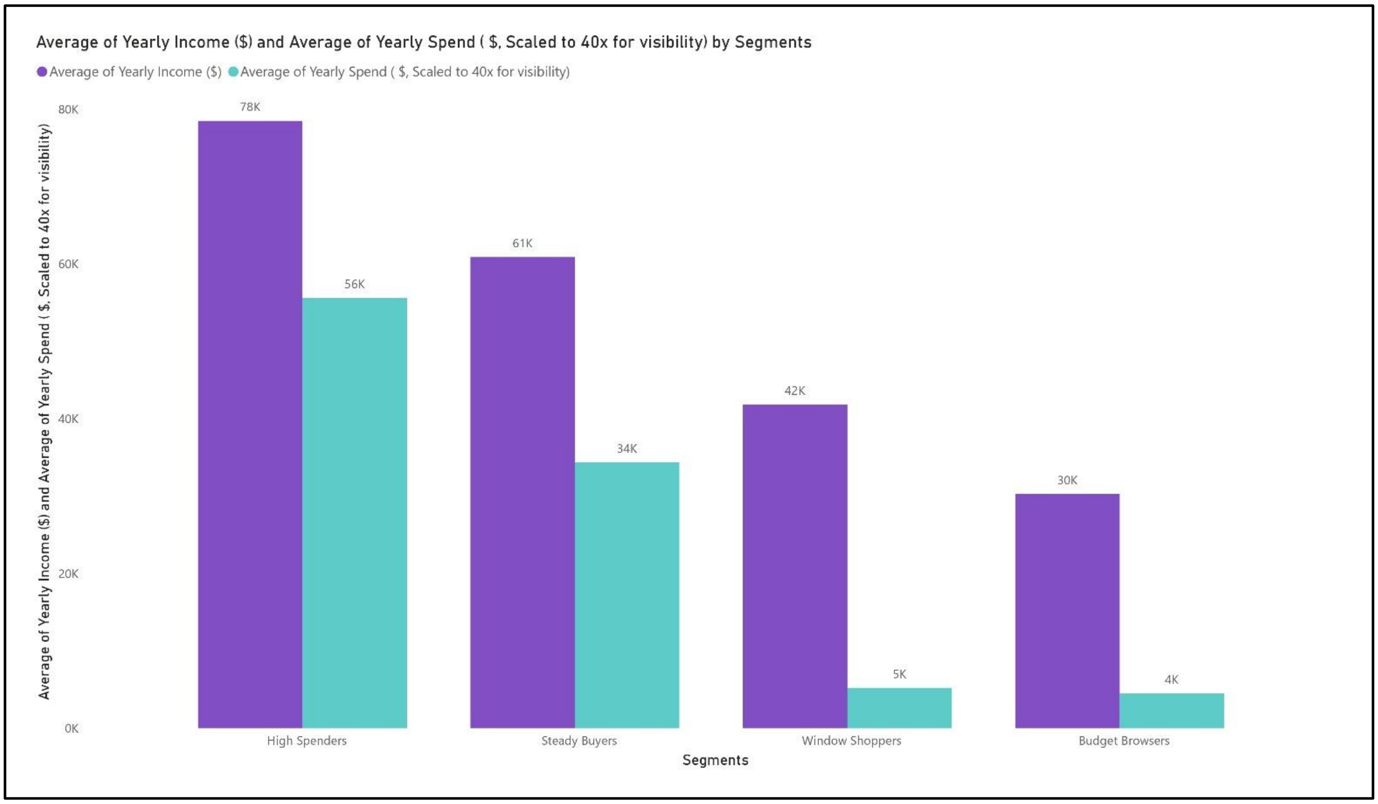

Figure 1 — High Spenders earn the most and spend the most Budget Browsers earn the least and spend the least. But income alone doesn't explain the browsing patterns below.

WHAT THIS ACTUALLY MEANS

BROWSING AND BUYING ARE NOT THE SAME THING

Window Shoppers and Budget Browsers visit the website roughly 3x more often than High Spenders (6.0–6.9 visits per month versus 2.3), yet they spend a fraction of the amount; as little as $112–130 a year compared to $1,390. For a large share of customers, browsing looks less like shopping and more like window shopping in the traditional sense: looking because it's enjoyable, or to compare prices, without any real plan to buy right now. This pattern is consistent with established consumer research on impulse and non-impulse buying behaviour (Rook, 1987).

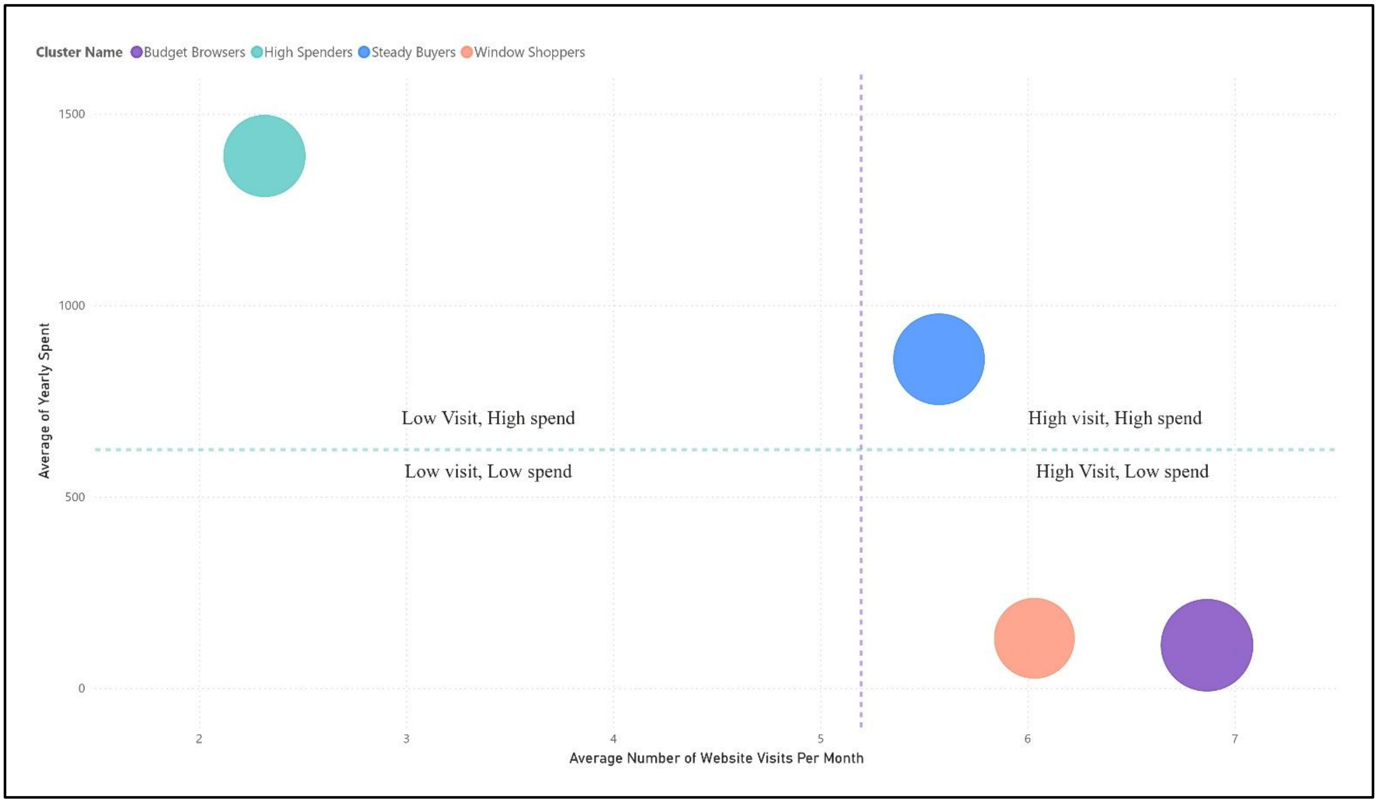

Figure 2 — High Spenders sit alone in the low-visit, high-spend quadrant, while Window Shoppers, Steady Buyers, and Budget Browsers all cluster toward high visits and low-to-moderate spend. Browsing frequency and spend move in opposite directions across every segment except one.

AGE AND FAMILY SITUATION CHANGE BUYING BEHAVIOUR

Window Shoppers are the oldest group on average and have the most children. This combination of older age, more kids and tighter household budgets leads to heavy browsing and restrained spending, though the clustering method here can only establish that these traits travel together, not that one causes the other. Price-conscious shopping behaviour of this kind, where consumers actively seek out the best deal before committing, is a well-documented pattern in consumer research (Lichtenstein, Ridgway, & Netemeyer, 1993). Budget Browsers, by contrast, are the youngest group, which may point to a simpler explanation: less disposable income at this life stage, rather than deliberate price comparison.

HIGH SPENDERS DON'T NEED TO BE CONVINCED

High Spenders barely visit the website compared to other groups, yet spend the most by a wide margin. This points to fast, habitual buying rather than deliberation, consistent with dual-process accounts of decision-making, where familiar, low-risk purchases are handled by fast, automatic thinking rather than slow, effortful evaluation (Kahneman, 2011). This may reflect greater product familiarity or habitual purchasing patterns built up over repeat visits. Practically, this group likely doesn't need much persuading, rather they need to be kept satisfied so they keep buying automatically.

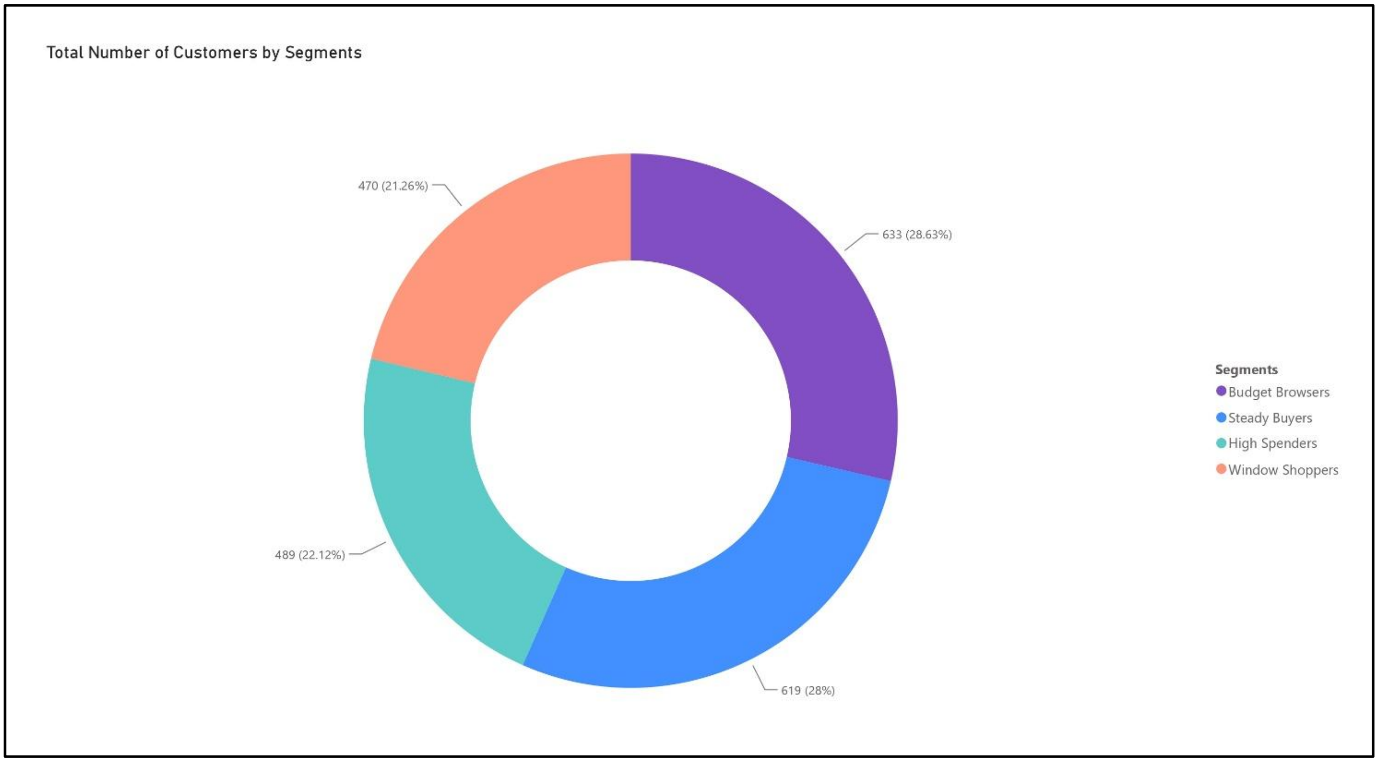

Figure 3 — Budget Browsers and Steady Buyers together make up over half the customer base (56.6%).

WHAT TO ACTUALLY DO WITH THIS

- Send discount nudges and reminder emails to Window Shoppers and Budget Browsers. They're already browsing, they just need a push.
- Don't over-invest marketing spend trying to convince High Spenders, retain and reward them instead; they're already buying.
- Investigate why browsing doesn't convert for over half the customer base- that gap is the real open question, and likely a psychological one, not just a pricing one.

AN OPEN QUESTION: COULD PERSONALISATION CLOSE THE GAP?

The segments above are static, they describe who the customers are but not what a retailer could do differently for each one in the moment. A natural next question in this day and age is whether AI-driven personalisation could close the browsing-buying gap for Window Shoppers and Budget Browsers: for example, by adjusting the depth of product recommendations or the timing of discount prompts based on browsing behaviour rather than treating all browsers alike. The evidence here is mixed. Personalisation can increase perceived usefulness and purchase intent, but only when it is paired with sufficient consumer trust in the retailer. When personalisation feels too closely tailored from a source the consumer does not yet trust, it tends to backfire into a sense of manipulation or lost autonomy rather than helpfulness (Bleier & Eisenbeiss, 2015). This suggests any personalisation strategy built on these segments would need to be tested for its effect on trust and perceived autonomy, not just conversion, before being rolled out, a question this dataset cannot answer on its own, but one worth investigating experimentally.

This analysis is based on one dataset, from one company, at one point in time. The patterns found here, particularly the browsing-buying gap are a strong signal worth investigating further, but they may not hold for every business, product category, or market. Understanding why the gap exists psychologically, rather than just confirming that it exists statistically, is the natural next step.

CONCLUSION

This analysis was set out to test a simple assumption, that browsing predicts buying, but after the analysis this was found out to be the opposite for a majority of customers. Four consistent segments emerged from the data: High Spenders, who buy with little browsing; Window Shoppers and Budget Browsers, who browse heavily without converting; and Steady Buyers, who sit in between. Age, income, and family size account for part of this pattern, but not all of it, which suggests the gap between browsing and buying is not purely a demographic or pricing story. Closing that gap is likely to require different tactics for different segments rather than one common strategy. Whether AI-driven personalisation can close it responsibly, without undermining the consumer trust it depends on, is the open question this paper leaves for further research.

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Dataset: Romero-Hernandez, O. (n.d). *Customer personality analysis* [Data set]. Kaggle. Full code and dashboard available in the linked GitHub repository.